

Analysis and interpretation of magnetotelluric data in characterization of geothermal resource in Eburru geothermal field, Kenya

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ABSTRACT

The magnetotelluric method (MT) is an essential geophysical method for the exploration of geothermal systems. In this study, the MT method was used to assess the extent of the geothermal resource in Eburru geothermal field, Kenya, with the aim of delineating the electrical conductivity structure of the area. Dimensionality analyses demonstrated that the MT data could be interpreted using two-dimensional approaches, but some localized 3-D effects were detected. A 2-D MT inversion was performed to generate resistivity models of Eburru geothermal field. Given its ability to recover complex resistivity models for the ground, three dimensional (3-D) MT inversion was also carried out, and a joint interpretation made from the 2-D and 3-D models. Both inversion approaches gave similar results and revealed a low resistivity layer ($< 10 \Omega\text{m}$) interpreted as clay cap, and an intermediate resistivity beneath interpreted as a geothermal reservoir immediately below the low resistivity. The sequence here infers the presence of geological structures controlling the geothermal system. The resistivity profiles analyzed revealed a structure of low resistivity ($< 10 \Omega\text{m}$) interpreted as the fluid pathway. This structure trend an S-N direction which is consistent with the faults orientation in the field and serves as a conducting channel for transporting heat from the heat source to the shallow region approximately 2 km above sea level.

1. Introduction

Geothermal energy is an alternative source of energy often found in remote areas where there is no other source of energy. The main advantage of geothermal source compared to other renewable sources like hydropower, solar and wind is that it does not depend on weather conditions, it is reliable and renewable. Geothermal power generation is often found in volcanic areas. East African Rift System (EARS), is one of the regions privileged with a remarkable geothermal potential (Endeshaw, 1988). Our region of interest is the Eburru field which is one of the geothermal areas in the Kenya rift.

Eburru is a Quaternary volcano located on the southern segment of Kenyan Rift, an arm of the larger EARS. It forms the highest topography within the entire Kenyan Rift at an elevation of 2800 m, while Badlands lies at a lower altitude to the North of Eburru massif towards Lake Elementaita (Fig. 1). The geothermal manifestations in Eburru include fumaroles, hot spring and altered grounds, a series of faults and fracture network system with general N-S trends and minor E-W trending faults. Therefore, it is important to investigate the geothermal system in order to delineate features related to the cap rock, reservoir, and heat source

as well as geological structures that may control the system.

Geothermal-water-rich rocks usually have relatively lower resistivity than original rocks, and the difference in the resistivity could be related to the temperature, water saturation, and degree of mineralization (Spichak et al., 2007).

Hydrothermal fluid-filled fractures usually have lower resistivity. The resultant low-resistivity anomalies have been the primary target for the geophysical exploration of geothermal resources (Simpson and Bahr, 2005). The MT method is a widely used resistivity method due to its high sensitivity to detect resistivity contrasts that characterize the subsurface structure of a geothermal system. MT data are inherently sensitive to the presence of fluids, both aqueous and partial melt, which weaken the crustal material (Zhang et al., 2015). The interpretation of Eburru MT data will take into consideration the structures, the geological setting and the alteration pattern that might influence the area's resistivity distribution. The results from surface geological mapping, gravity, and geothermal exploration wells will be used as constraints to model the MT data (Simiyu and Keller, 2001).

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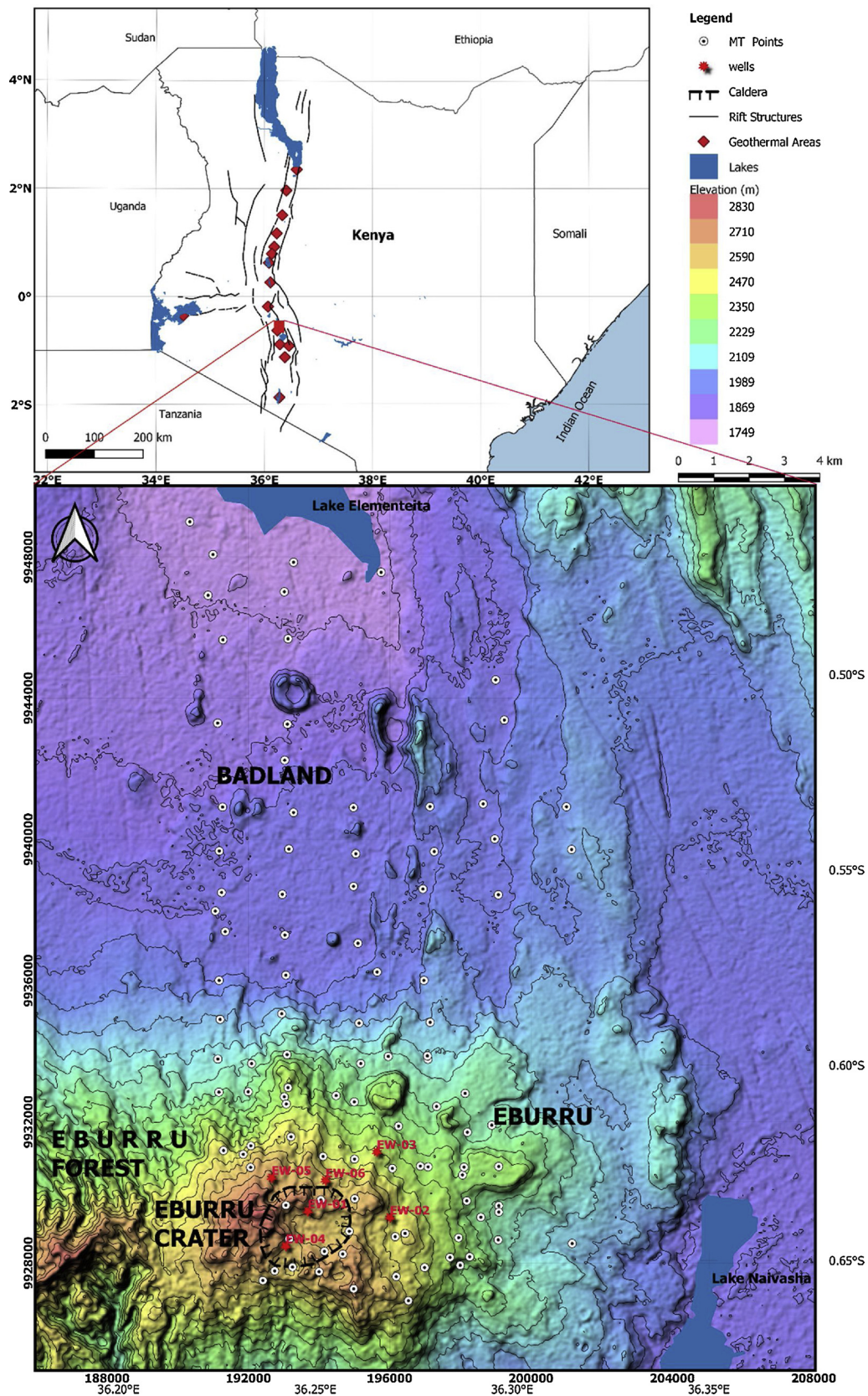


Fig. 1. Map showing elevation and MT site locations in the survey area of Eburru Geothermal Field. Red stars denote the location of wells. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

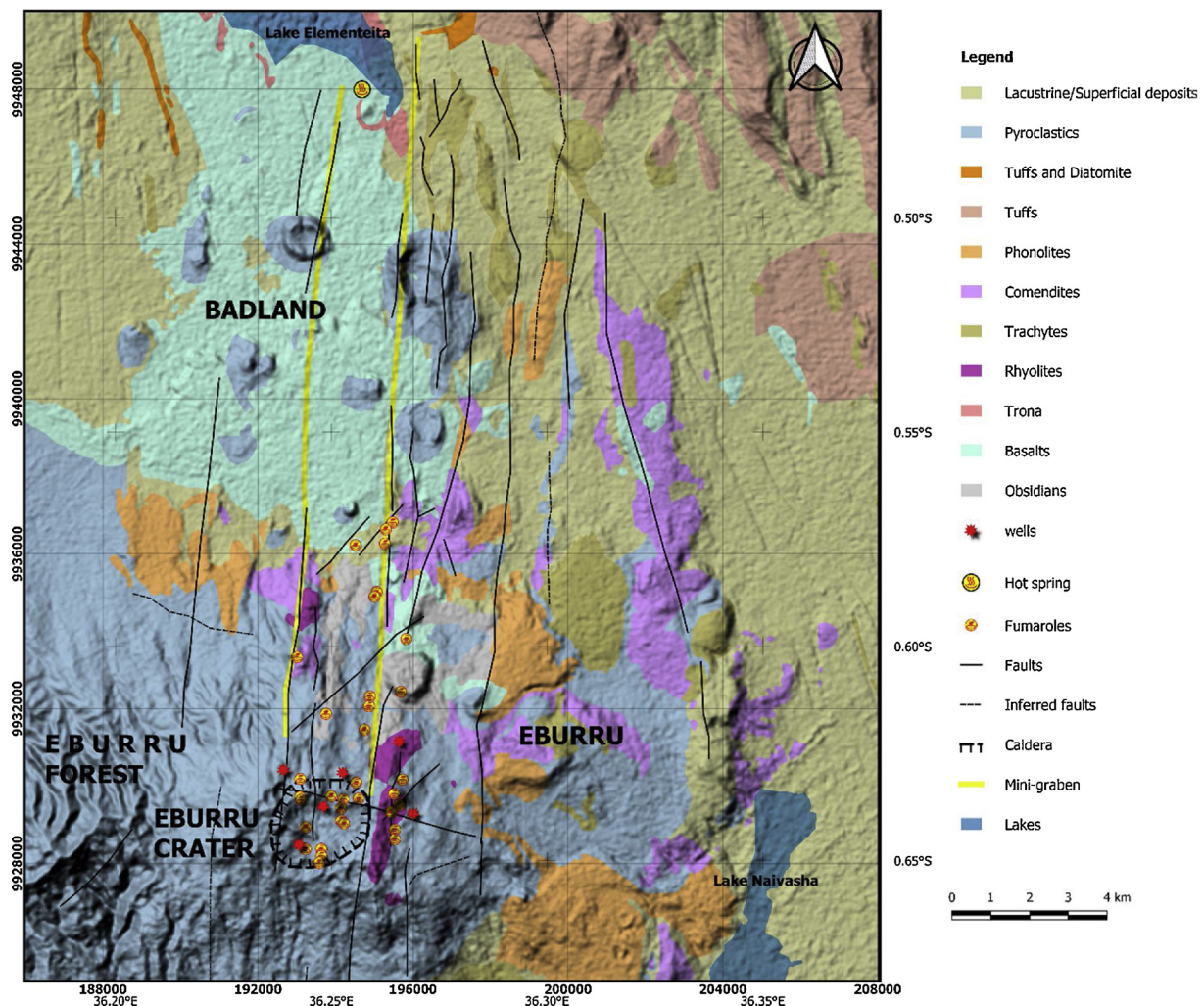


Fig. 2. Geological map of the Eburru geothermal field (Modified from Thompson and Dodson, 1963).

1.1. Geological setting

The Eburru geothermal field consists of east and west volcanic centers which are composed of rhyolites, obsidians, phonolites, pyroclastics, comendites, tuffs, basalts of upper Pleistocene, trachytes, and pumice (Lagat, 2003). The two volcanic centers extends as far to the west as the Mau escarpment and are arranged in an E-W trend. Pyroclastics deposits are widely spread in the western part of the Eburru crater within the Eburru forest area. The basalts which are of recent age are not masked by vegetation and covers the badland region located in the northern side of the field. The Eburru field is characterized by numerous faults that trend in N-S direction and by caldera-like features (Fig. 2). These faults are of two categories, the older rift faults, and the young faults. The older rift faults appear to trend in an NW-SE direction while the young faults observed as open fissures are oriented in the N-S direction. The faults to the west are downthrown to the east while those to the east are downthrown to the west, hence leading to the formation of a graben structure (Simiyu, 1990). The eastern part of the Eburru crater is characterized by large open fractures and numerous faults marked by steaming grounds and fumaroles. These faults appear to host the crater and might have served as a conduit of lava flows during their ascent to the surface. The faults could also form fluid pathways and reservoir for geothermal fluids (Beltran, 2003; Muchemi, 1990; Omenda and Karingithi, 1993; Velador et al., 2003). Most fumaroles align along the NNE-SW, E-W and near the N-S younger faults. Calcite, silica, sulfur and green and red-brown clay deposition characterize most

of the fumaroles in the Badlands area. The MT profiles presented in this paper are focused on the central region of the Eburru field (Fig. 10). Within this region, there are numerous S-N trending faults, active fumaroles, and the drilled exploration wells.

1.2. Previous geophysical studies

A number of geophysical studies have been conducted in the central Kenya rift with more focus in Olkaria (Bodvarsson et al., 1987; Ndombi, 1981; Simiyu and Keller, 2000, 2001; Simiyu and Malin, 2000) and Menengai (Kanda et al., 2019; Mariita and Keller, 2007; Omenda and Simiyu, 2015; Saitet et al., 2016; Simiyu, 2010, 2013; Wamalwa et al., 2013) geothermal fields. However, only a few geophysical studies have been carried out in Eburru geothermal field. A gravity survey by Simiyu (1990) revealed an N-S axial high along a graben, most likely caused by dense intrusives along N-S fault zones and at major structural intersections within the area. He suggested that the fluid feeding the geothermal system is flowing along vertical conduits and the heat emanates from narrow intrusive bodies found at fault junctions. MT and TEM surveys revealed a detailed image of the resistivity pattern in the field. A TEM cross-section showed a low resistivity zone at shallow depth overlaying a high resistive plume which can be attributed to low-temperature hydrothermal minerals (Wameyo, 2007). Results from 1D joint inversion of MT and TEM Eburru data revealed a detailed image of the resistivity pattern with a very thin layer of high near-surface resistivity ($> 60 \Omega\text{m}$) covering almost the entire area. This layer overlays a

uniform near-surface low-resistivity layer ($< 10 \Omega\text{m}$) resulting from the geothermal activity, such as fumaroles (Omiti, 2014). He suggested the existence of a deep conductive body ($< 10 \Omega\text{m}$) beneath a resistive zone in the southwest of the field indicating a probable heat source of Eburru geothermal system.

1.3. MT method

The magnetotelluric (MT) technique is a passive electromagnetic (EM) technique that involves measuring fluctuations in the natural electric, E , and magnetic, H , fields in orthogonal directions at the surface of the Earth as a means of determining the conductivity distribution of the Earth at depths ranging from some metres to several hundreds of kilometres. Natural sources of MT fields above about 1 Hz are thunderstorms worldwide, from which lightning radiates fields which propagate to great distances. At frequencies below 1 Hz, the bulk of the signal is due to current systems in the magnetosphere set up by solar activity. The basic MT response, Z , can be defined as the linear relationship between horizontal electric and magnetic field variations at a specific station at the Earth's surface. It is complex-valued, which relates the fields in the frequency domain.

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \frac{1}{\mu_0} \begin{pmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix} \quad (1)$$

Where E_i is the electric field, and B_i ($i \in [x, y]$) is the magnetic field variations observed along the north (x) and east (y) directions. The μ_0 is the magnetic permeability of free space. Z , also known as the impedance tensor, is frequency dependent and contains the information about the subsurface conductivity structure. The tensor elements Z_{ij} are usually displayed in terms of the apparent resistivity, ρ_a

$$\rho_{aij} = \frac{1}{\mu_0 \omega} |Z_{ij}(\omega)|^2 \quad (2)$$

where ω is the angular frequency and the phase angle (Φ) can be written as:

$$\Phi_{ij} = \arctan\left(\frac{\text{Im}(Z_{ij}(\omega))}{\text{Re}(Z_{ij}(\omega))}\right) \quad (3)$$

For a comprehensive introduction into MT much information has been given by Simpson and Bahr (2005), Berdichevsky and Dmitriev (2008) or Chave and Jones (2012).

Being a tensor, Z also contains information about dimensionality and direction. For a 1-D Earth, wherein conductivity varies only with depth, the diagonal elements of the impedance tensor, Z_{xx} , and Z_{yy} (which couple parallel electric and magnetic field components) are zero, whilst the off-diagonal components (which couple orthogonal electric and magnetic field components) are equal in magnitude, but have opposite signs, i.e.

$$\left. \begin{aligned} Z_{xx} &= Z_{yy} = 0 \\ Z_{xy} &= -Z_{yx} \end{aligned} \right\} 1\text{-D} \quad (4)$$

For a 2-D Earth, in which conductivity varies along one horizontal direction as well as with depth, Z_{xx} and Z_{yy} are equal in magnitude but have opposite sign, while Z_{xy} and Z_{yx} differ, i.e.

$$\left. \begin{aligned} Z_{xx} &= -Z_{yy} \\ Z_{xy} &\neq -Z_{yx} \end{aligned} \right\} 2\text{-D} \quad (5)$$

For a 2-D Earth with the x - or y -direction aligned along electromagnetic strike, Z_{xx} and Z_{yy} are again zero. Mathematically, a 1-D anisotropic Earth is equivalent to a 2-D Earth.

With measured data, it is often not possible to find a direction in which the condition that $Z_{xx} = Z_{yy} = 0$ is satisfied and this may be due to distortion or to 3-D induction (or both). Generally, the dimensionality evinced by data is scale dependent (Simpson and Bahr, 2005).

1.4. MT survey and data acquisition

MT data were collected in three field campaigns by Kenya Electricity Generating Company (KenGen) staff in 2006, 2013 and 2016 using Phoenix MTU-5 instruments. A total of 107 soundings white dots shown in Fig. 1 were used in this study. A remote reference site was based in Olkaria 30 km from Eburru field in order to remove electromagnetic noise from the electromagnetic signals at each measuring station. The electric field in the x -direction (E_x) was measured in the north-south direction, and the field in the y -direction (E_y) was recorded in the east-west direction. The distance between the electrodes was approximately 60 m. The magnetic component H_x describes the north-south direction, H_y the east-west direction, and H_z into the ground surface. All five components of the time-varying electromagnetic field (E_x , E_y , H_x , H_y , H_z) were recorded at the local station and the remote station. At each station, the total variation of electric in the magnetic field was recorded for about 18 h to collect data for long periods and also take advantage of the strong signals usually available in the late hours of the night. The above raw data were processed by using SSMT2000 software manufactured by Phoenix Geophysics to transform them into frequency domain expression in terms of apparent resistivities and phase of impedance as a function of frequency for each station. The time series data were Fourier transformed then processed and edited in the MT Editor program.

2. Data analyses

2.1. Data imaging

MT sounding curves can provide information about the structures that might be expected from subsequent modeling and inversion. Fig. 3 shows MT sounding curves of TE and TM modes obtained from the 107 sites in Eburru geothermal field. The apparent resistivity of both modes shows high values at low periods (< 0.01 s) and decreases as the period increases to below 1 s. Subsequently, the apparent resistivity increases up to a period of 10 s before it falls to a period of 100 s.

2.2. Dimensionality analysis

Dimensionality analysis of MT data is a common procedure for inferring the main properties of the subsurface geoelectric structures such as the strike direction or the presence of superficial distorting bodies, and enables the most appropriate modeling approach to be determined (Martí et al., 2010). MT data analyses are done to derive a dataset suitable for defining the resistivity model of the Earth from the observed MT data. It is therefore essential to investigate the dimensionality of the MT responses and determine if the data require a 1-D layered Earth, 2-D model invariant in the strike direction or 3-D resistivity model. Phase tensor (Caldwell et al., 2004) was used to check the dimensionality of the Eburru MT data.

2.3. Phase tensor analysis

The phase tensor (Booker, 2014; Caldwell et al., 2004) is a useful tool in assessing the dimensionality of MT impedance data and is defined as the product of an inverse real matrix and an imaginary matrix of the impedance tensor. It can be graphically represented as an ellipse defined by three values: the maximum phase Φ_{max} , the minimum phase Φ_{min} , and the skew angle β (Fig. 4). Φ_{max} and Φ_{min} correspond to the major and minor axes of the ellipse with their directions indicating the two orthogonal electrical principal axes, or two possible strike directions. Under 1-D conditions, Φ_{max} and Φ_{min} are equal, and the ellipse is a circle. The skew angle β gives a measure of the dimensionality and defined by Caldwell et al. (2004). The skew is zero for a 1-D or 2-D resistivity structures, and non-zero values indicate a 3-D resistivity structure.

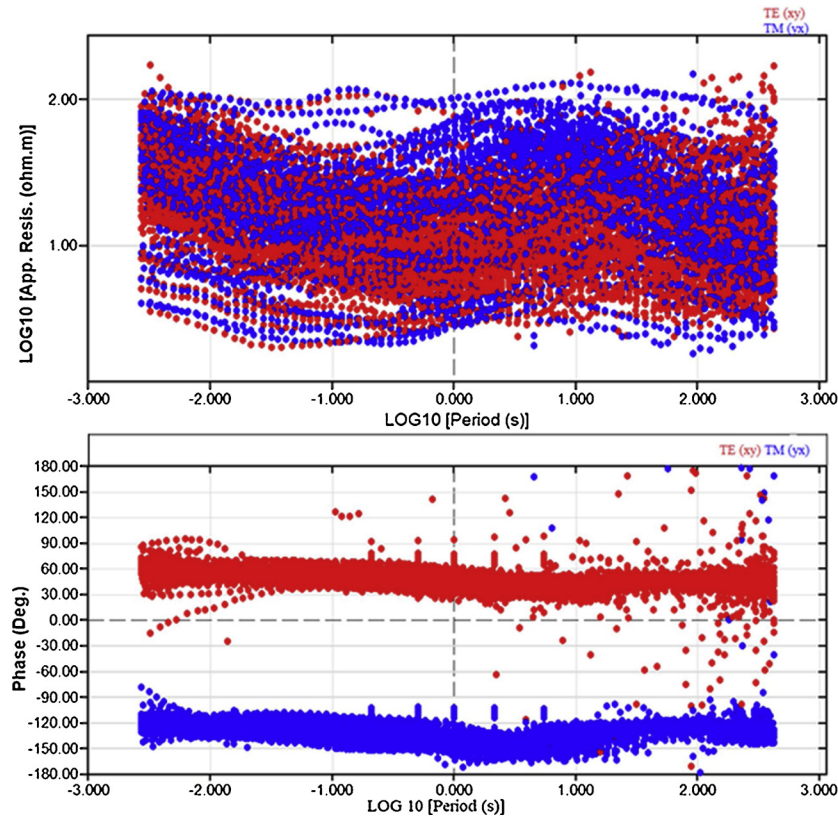


Fig. 3. Plots of all MT sounding curves of TE and TM modes from the 107 stations in Eburru field. The red dots indicate the TE mode and the blue dots indicate TM mode (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

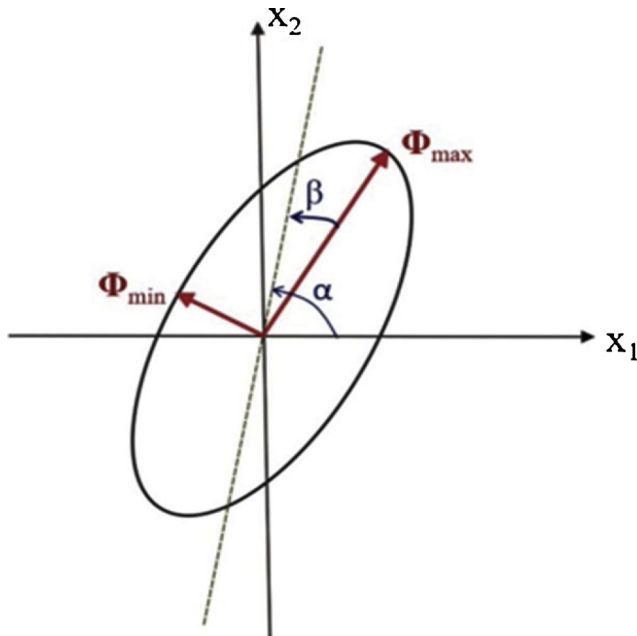


Fig. 4. The phase tensor plotted graphically as an ellipse. The $\Phi_{\max}$ and $\Phi_{\min}$, respectively represent the maximum and minimum principal axes. If the phase tensor is not N-Symmetric, a third coordinate invariant is needed to characterize the tensor: the skew angle, β . The angle $\alpha - \beta$, which gives the orientation of the major axis of the ellipse, defines the relationship between the tensor and the observational reference frame (X_1 and X_2) from Caldwell et al. (2004).

The phase tensor maps for Eburru geothermal field at different periods are shown in Figs. 5–8 and reveal the dimensionality of the MT data. These periods were chosen since the development of 3-D behavior

would be shown more clearly and the period range covers the reservoir better. Most of the soundings showed relatively small skew angles at short periods. In Figs. 5–7 many stations exhibit one-dimensionality as indicated by the phase tensor plotting as a circle as is nearly the case for some of the phase tensors at short periods. However, at longer periods some stations start to exhibit complex aspects, which do not support the presence of a 2-D resistivity structure ($\beta > 3$), as shown in blue and red colors, see Fig. 8. The phase tensor analysis shows 1-D and 2-D dimensionality at short periods as the majority of the stations are characterized by small skew angle and detects 3-D effects at long periods characterized by high skew angles.

The results from the dimensionality analysis show a shallow 1-D, 2-D features and deeper 3-D features.

2.4. Static shift correction

Static shift is a possible source of distortion caused by resistivity inhomogeneity close to the electric dipoles. The static shift problem was addressed by use of a spatial median filter method with a diameter of 3 km applied to sites that showed a significant split in the apparent resistivity curves at high frequency. Also, the ModEM programme accounts for the remaining static shift effects by introducing a scattered conductivity distribution in the near-surface layers (Meqbel et al., 2014; Tietze and Ritter, 2013).

3. MT data inversion

3.1. Two-dimensional (2-D) inversion

In a 2-D inversion, it is assumed that the resistivity can vary with depth and in one lateral direction and that the resistivity is constant in the other horizontal direction (electrical strike). The inversion is done for soundings on profiles which are roughly perpendicular to the

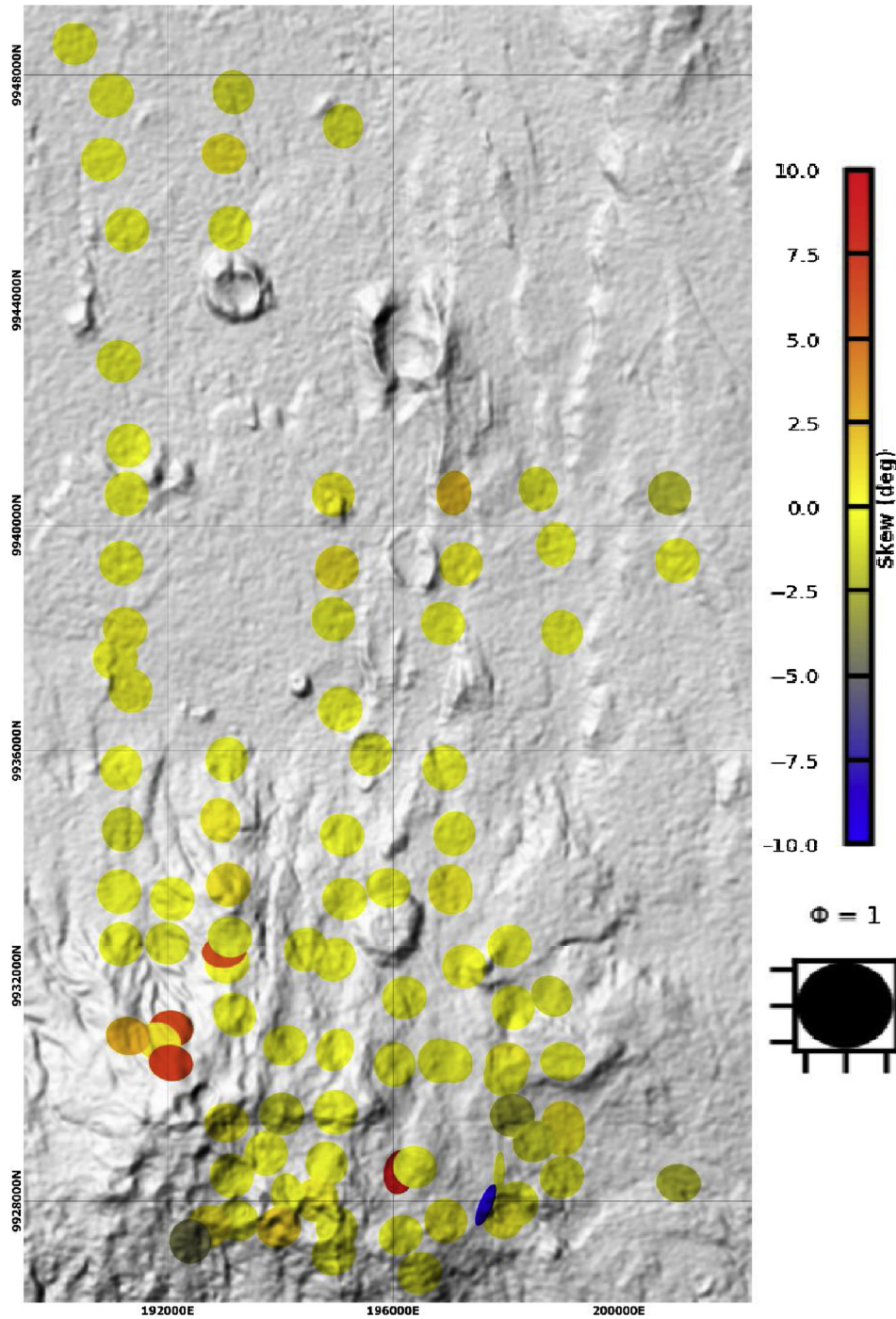


Fig. 5. Phase tensor map for a period of 0.01 s.

electric strike. If the field setup coordinate system is not parallel and perpendicular to the electrical strike, the MT impedance tensor data are mathematically rotated with one axis perpendicular to the electrical strike and the other axis parallel to it. The inversion is then done to fit the apparent resistivity and phase data computed from the impedance elements with the electric field perpendicular to the strike (called the TM-mode) or the mode with the field parallel to the strike (called the TE-mode), or both simultaneously.

For 2-D inversion of Eburru MT data, we used Occam's 2-D inversion code (version 3.0) developed by Scripps Institution of Oceanography based on [deGroot-Hedlin and Constable \(1990\)](#).

Occam's 2-D inversion is based on the minimization of the following unconstrained functional:

$$U = \|\partial_y m\|^2 + \|\partial_z m\|^2 + \mu^{-1} \{ \|W(d - F(m))\|^2 - X_n^2 \} \quad (6)$$

where the expression, $\|\partial_y m\|^2 + \|\partial_z m\|^2$ is a norm of the model roughness, μ^{-1} is the Lagrange multiplier, the third term in the equation represents the data misfit, W is $M \times M$ diagonal weighting matrix, d represents the observation vector, and $F(m)$ represents the model response.

3.2. Transverse electric (TE) mode and transverse magnetic (TM) mode

The 2-D magnetotelluric field consists of the TE and TM modes. The TE mode is related to the E-polarized wave generating the longitudinal MT curves (telluric current flows along the structures), and the TM

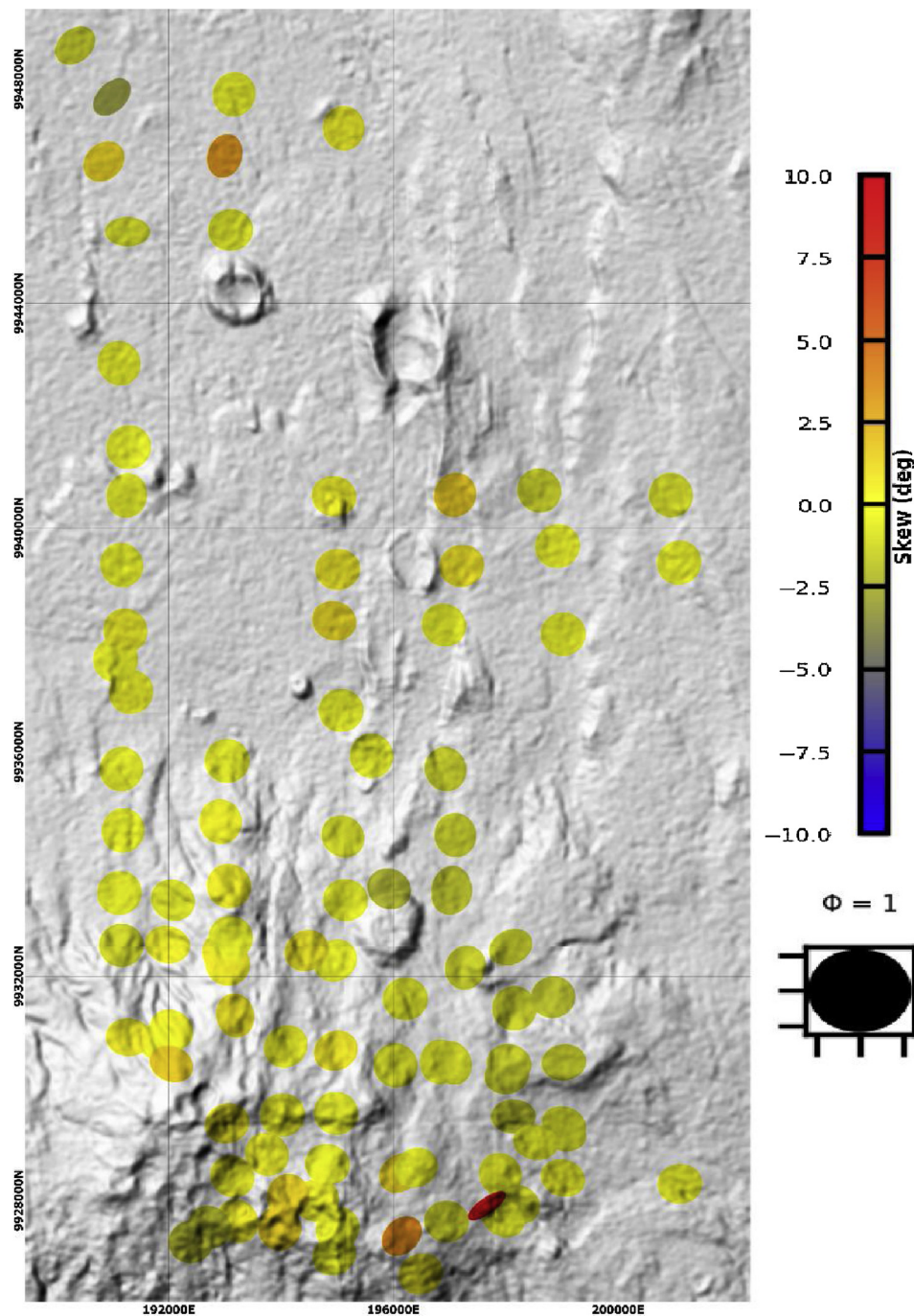


Fig. 6. Phase tensor map for a period of 0.1 s.

mode is related to the H-polarized wave generating the transverse MT curves (telluric current flows across the structures). The TM and TE modes offer different sensitivities to near- surface and deep structures and provide different accuracies of 2-D approximation of real 3-D bodies. The mode most unaffected by 3-D effects depends on the position of the 3-D structure with respect to the regional 2-D strike direction. When the 3-D body is normal to the regional strike, the TE-mode is affected mainly by galvanic effects, while the TM-mode is affected by galvanic and inductive effects. In this case, a 2-D interpretation of the TM-mode is prone to error. When the 3-D body is parallel to the regional 2-D strike, the TE-mode is affected by galvanic and inductive effects and the TM-mode is affected mainly by galvanic effects, making it more suitable for 2-D interpretation (Wannamaker et al., 1984). Thirteen parallel profiles cutting across the geologic structures were

selected for the 2-D inversion. A joint 2-D inversion, which includes both TE and TM mode was performed along all the profiles. The joint inversion of the TE and TM mode data was performed in order to derive an overall picture of the subsurface conductivity structure in Eburru geothermal field that would explain the data from both polarizations simultaneously.

For 2-D models, the pseudosection plot of observed and predicted responses were generated. For Profile 1 the pseudosection plot for TE and TM modes for both observed and predicted in Fig. 9 shows a good fit. The TM mode shows a relatively better fit than the TE modes as it shows a small residual for Profile 1.

For a 2-D model, the earth can be described using a grid of rectangular prisms, each having a uniform conductivity. The grid is terminated horizontally by uniform layers and below by prisms extended

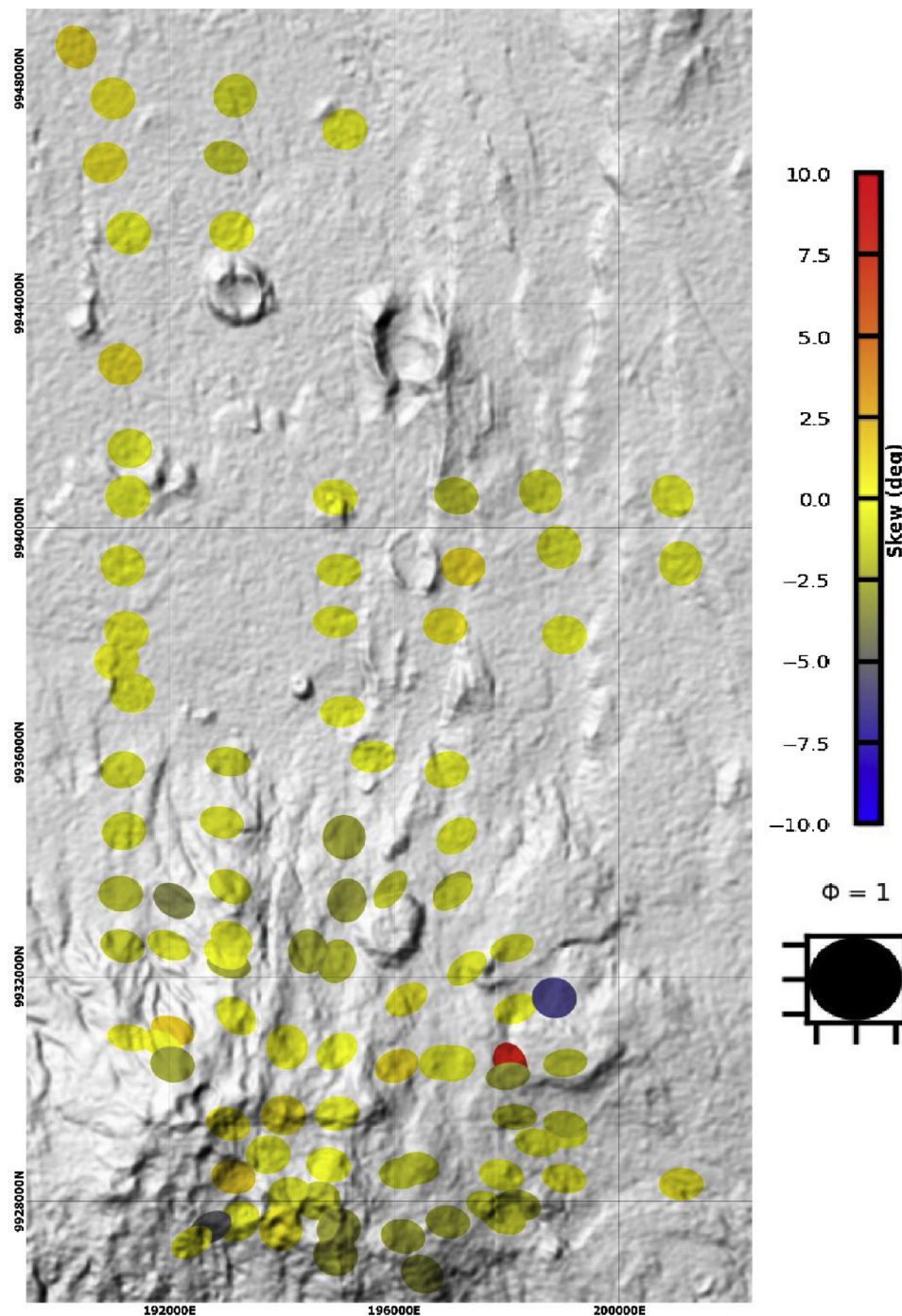


Fig. 7. Phase tensor map for a period of 1.0 s.

with depth. This grid is known as the regularization mesh. The separate blocks are made smaller than the data resolution length so that the positions of the block borders do not affect the final model. For the 2-D inversion, a mesh was generated for each profile with a block width set to 50 m. Three blocks were added to the mesh boundaries in both directions, and the dimensions of these boundary blocks increased with a factor of 2. In the vertical direction, a total of 70 layers were used with the first-layer thickness of 20 m increasing logarithmically for the subsequent layers. A homogeneous halfspace of 50 Ωm was used as the starting model for all 2-D inversions. To address any remaining static shifts that had not been corrected using the median method, an apparent resistivity error floor of 5% was used. Preferred models that fit the MT data with an overall root mean square (r.m.s) misfits of 0.72

(profile 1), 2.18 (profile 2), 1.43 (profile 3) and 1.51 (profile 4) are presented in Figs. 11–14 respectively. The choice of the four parallel profiles (Fig. 10) was made due to the peculiar geological settings in the study area.

3.3. Two-dimensional inversion results

The resistivity models obtained from the 2-D inversion schemes of the four resistivity profiles in Fig. 10 are shown in Figs. 11–14. The major resistivity features on the four profiles are similar. From the 2-D inversion model of Profile 1, two conductors (C1, C2) and two resistors of very high resistivity (R1, R2 > 300 Ωm) were identified as shown in Fig. 11. The most striking resistivity feature in this profile is a deep-

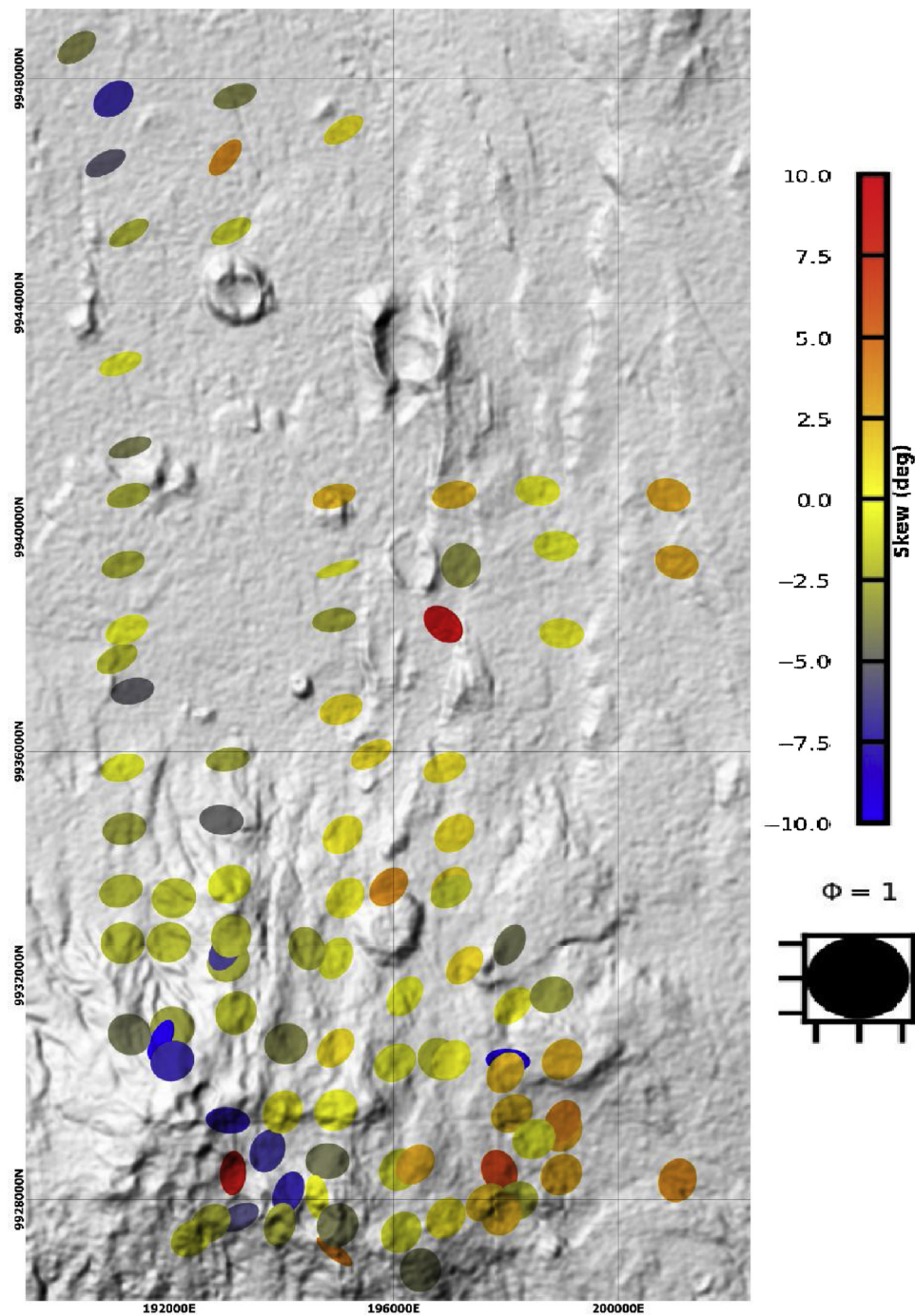


Fig. 8. Phase tensor map for a period of 10 s.

reaching conductive channel extending almost from the surface connecting C1 and C2 down to a depth of 5 km. At a depth of 1.2 km from the surface, the channel-like conductive body is observed to spread across the profile and extends deeper between sites E86 and E85. On profile 2 there appear to be also a deep conducting channel connecting conductor C3 and C4 that extends to a depth of 5 km. This profile has a large resistivity structure ($R3 > 300 \Omega\text{m}$) covering most of the profile and extends to the surface near site E02 (Fig. 12). Profile 3 shown in Fig. 13 presents similar trends as profile 1 and 2 with a deep conducting channel that stretches from the surface down to the bottom of the profile. Two resistors (R4 and R5) are seen on this profile which could suggest the presence of an impervious rock or a cold formation. On profile 4 (Fig. 14), there are pockets of conductive bodies near the

surface which appear to be connected to a conductive zone (C6) at a depth of 5 km. A resistive body (R6) is also observed on the east part of this profile. One similar structure noticed on these W-E trending resistivity profiles aligned in S-N direction is this resistive body on the east part. The deep conducting channel appears to be crossing all the four resistivity profiles and extends beyond profile 1 to the southern part of the field.

3.4. Three-dimensional (3-D) MT inversion

The objective of a 3-D MT inversion is to produce a 3-D resistivity model consistent with MT data that require it. The effective depth of investigation of the 3-D inversion is not limited by geometric

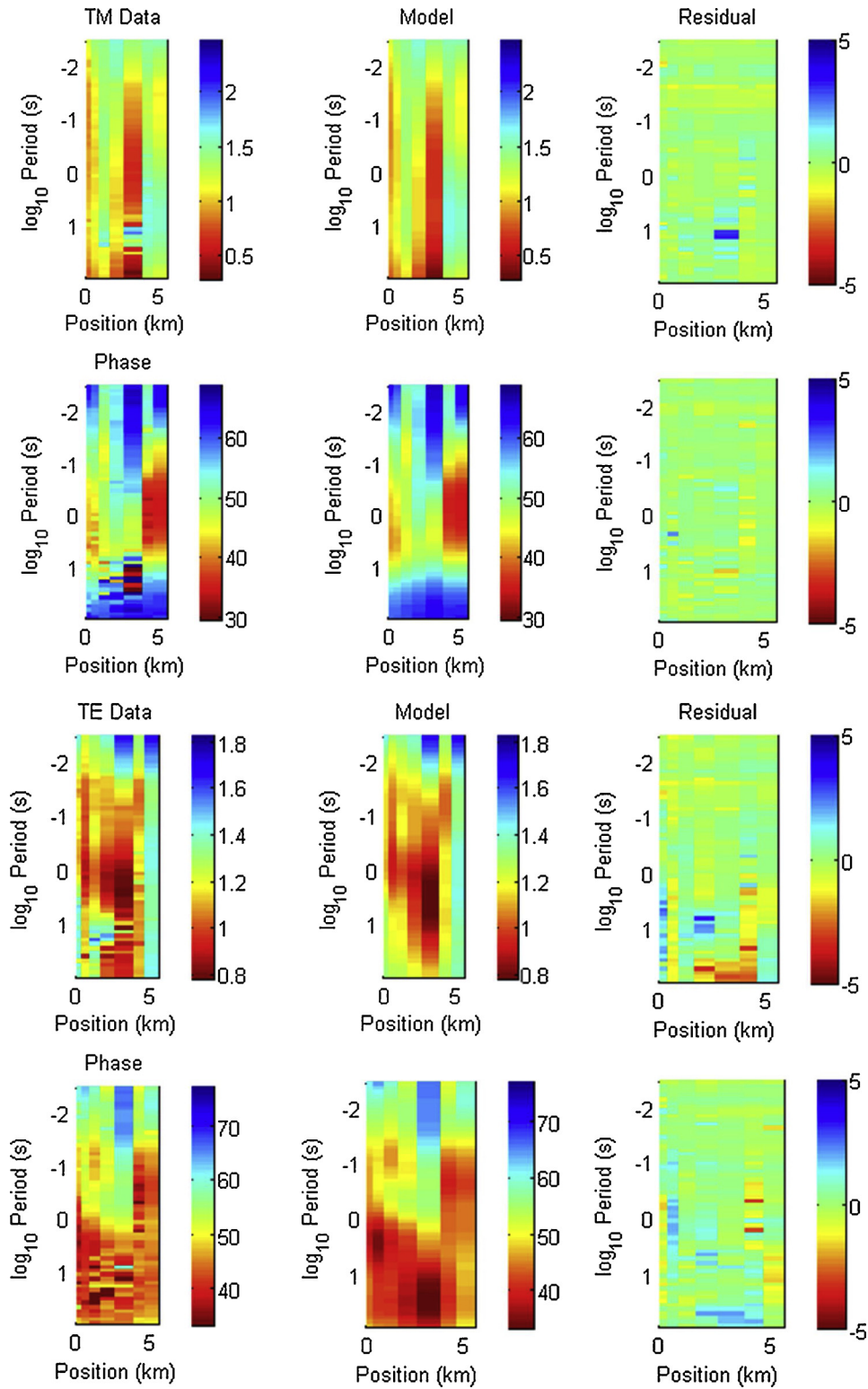


Fig. 9. Pseudosections of observed and calculated apparent resistivity and phase for TM (above) and TE mode (below) for Profile 1. The residual map indicates the differences between the observed and predicted responses.

assumptions as is the case with 1-D and 2-D inversion. However, the depth is constraint by the frequency range of the data used in the inversion, the aperture of MT station coverage, and the average resistivity of the model (Cumming and Mackie, 2010).

Three dimensional (3-D) MT inversion is widely used in geothermal exploration to image resistivity structures. Inversion codes such as WSINV3DMT and ModEM are freely available for academic purposes.

In this study, ModEM code developed by Egbert and Kelbert (2012), Meqbel (2009) and Kelbert et al. (2014) was used to analyze the MT data. ModEM code for 3-D inversion is based on the minimization of the following penalty functional:

$$U(\mathbf{m}, \mathbf{d}) = (\mathbf{d} - \mathbf{F}(\mathbf{m}))^T \mathbf{C}_d^{-1} (\mathbf{d} - \mathbf{F}(\mathbf{m})) + \lambda (\mathbf{m} - \mathbf{m}_0)^T \mathbf{C}_m^{-1} (\mathbf{m} - \mathbf{m}_0) \quad (7)$$

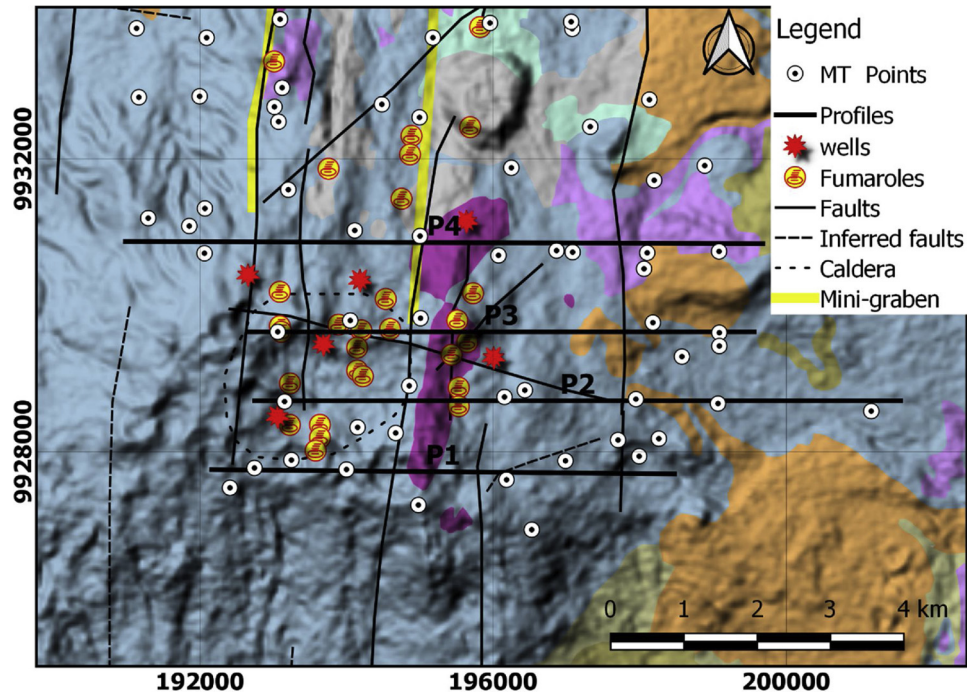


Fig. 10. Geological map showing MT profiles for 2-D inversion in Eburru geothermal field.

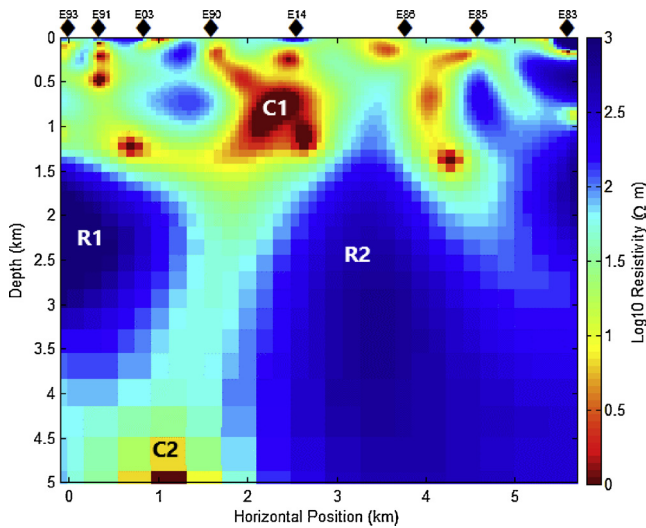


Fig. 11. 2-D inversion modeling of the MT observation points along Profile 1.

The first term in Eq. 7 above represents the data misfit between the measured ($\mathbf{d}$) and model response, $\mathbf{F}(\mathbf{m})$. The second term describes the model update between the estimated model ($\mathbf{m}$) and initial model ($\mathbf{m}_0$). $\mathbf{C}_d$ and $\mathbf{C}_m$ are the data and model covariances, respectively, λ is the regularization parameter. The initial model, $\mathbf{m}_0$ is updated iteratively by line search strategy. The 3-D forward scheme is based on finite differences, and the inverse scheme exploits the nonlinear conjugate gradient method. The 3-D inversion was applied to MT data obtained from 107 stations. The starting model was a uniform 50 Ωm half-space with 80, 53 and 50 cells in the x, y and z directions respectively, including x padding of 5 and y padding of 4. In the vertical direction, a total of 50 layers were used with a first layer thickness of 10 m increasing with a factor of 1.1 for the subsequent layers. The full impedance tensor was inverted in the frequency range of 0.00034 Hz–319.49 Hz for a total of 80 frequencies. The full impedance tensor and tipper were inverted to ensure that off-profile structures were resolved as reliably as possible (Siripunvaraporn et al., 2005). The

starting RMS of the inversion was 32.38, decreasing to 4.49 after 132 iterations. In general, the data misfit of the final model is satisfactory with limited over or under fitted data sections (Figs. 15–18).

3.5. Three-dimensional inversion results

From the 3-D modeling and inversion of MT data, two profiles passing through the same soundings used for 2-D inversion were analyzed for this study. The resistivity structures found on the western side of these profiles show similar characteristics, which might be attributed to their alignment. Profile 1 indicates the presence of two conductors C1, C2 and two resistors R1 and R2 (Fig. 19). Conductor C1 ($< 10 \Omega\text{m}$) is observed at an elevation of 2000 m a.s.l extending down to 1500 m a.s.l and appears to connect to a second conductive body C2. Conductor C2 with resistivity values of between 3 and 10 Ωm is located below C1 and extends from an elevation of 700 m a.s.l to ~2500 m b.s.l.

Resistor R1 ($> 80 \Omega\text{m}$) is confined within the two conductors while resistor R2 ($> 300 \Omega\text{m}$) extends from the surface almost to an elevation of ~2500 m b.s.l (Fig. 19). The second profile shown in Fig. 20 consists of three conductors C3, C4 and C5, which appear to enclose resistor R3 ($> 300 \Omega\text{m}$). Conductor C4 is located directly below a highly conductive C3 (~3 Ωm), and they extend from the near-surface to an elevation of ~2500 m b.s.l. The conductive zone C5 is found on the east part of the profile stretching from a lower elevation and extends to a higher elevation of 1200 m a.s.l.

3.6. Two-dimensional versus 3-D inversion modelling

Geological structures in Eburru geothermal field trend in the North to South direction, therefore the parallel profiles were selected from West to East direction to cut across the structures, and this was guided by the fact that these structures control the movement of fluid in a geothermal field. Horizontal cross sections were extracted from the final 3-D inversion model to compare with the 2-D inversion models. Both the 2-D and the 3-D resistivity models (see Figs. 19 and 20) show similar resistivity features. Electrical resistivity features C1, C2, C3, C4, R1, R2, and R3, were imaged by both inversion models, and generally, have consistent geometries. Therefore, the comparison of the 2-D and 3-

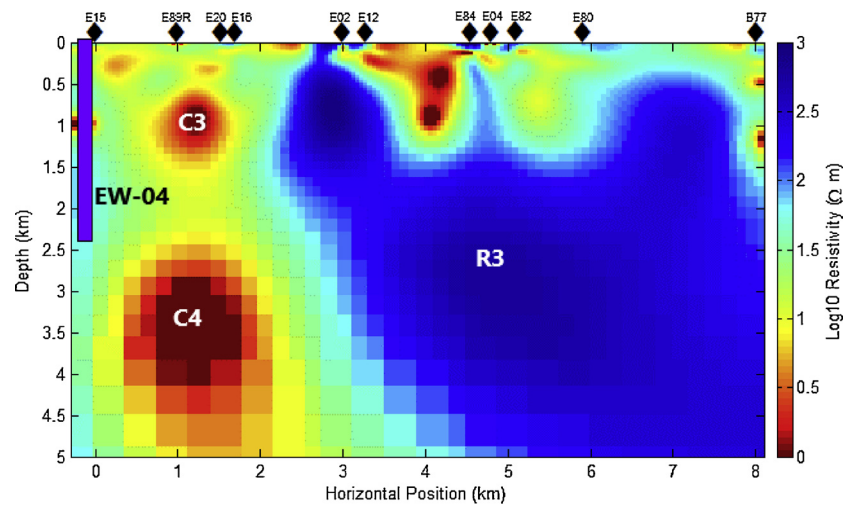


Fig. 12. 2-D inversion modeling of the MT observation points along Profile 2.

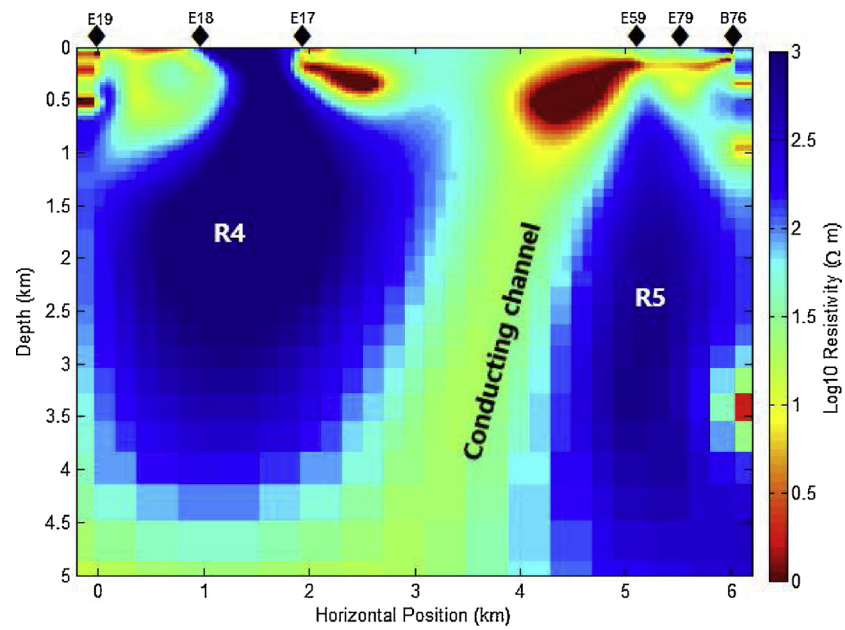


Fig. 13. 2-D inversion modeling of the MT observation points along Profile 3.

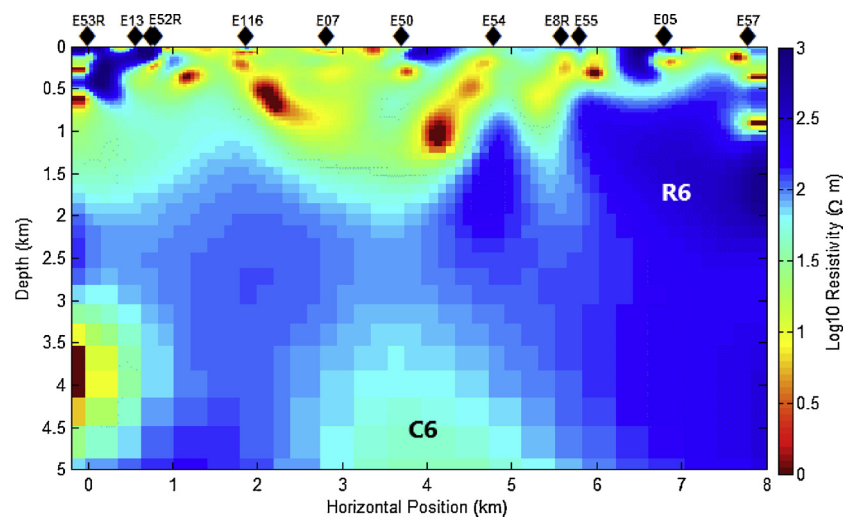


Fig. 14. 2-D inversion modeling of the MT observation points along Profile 4.

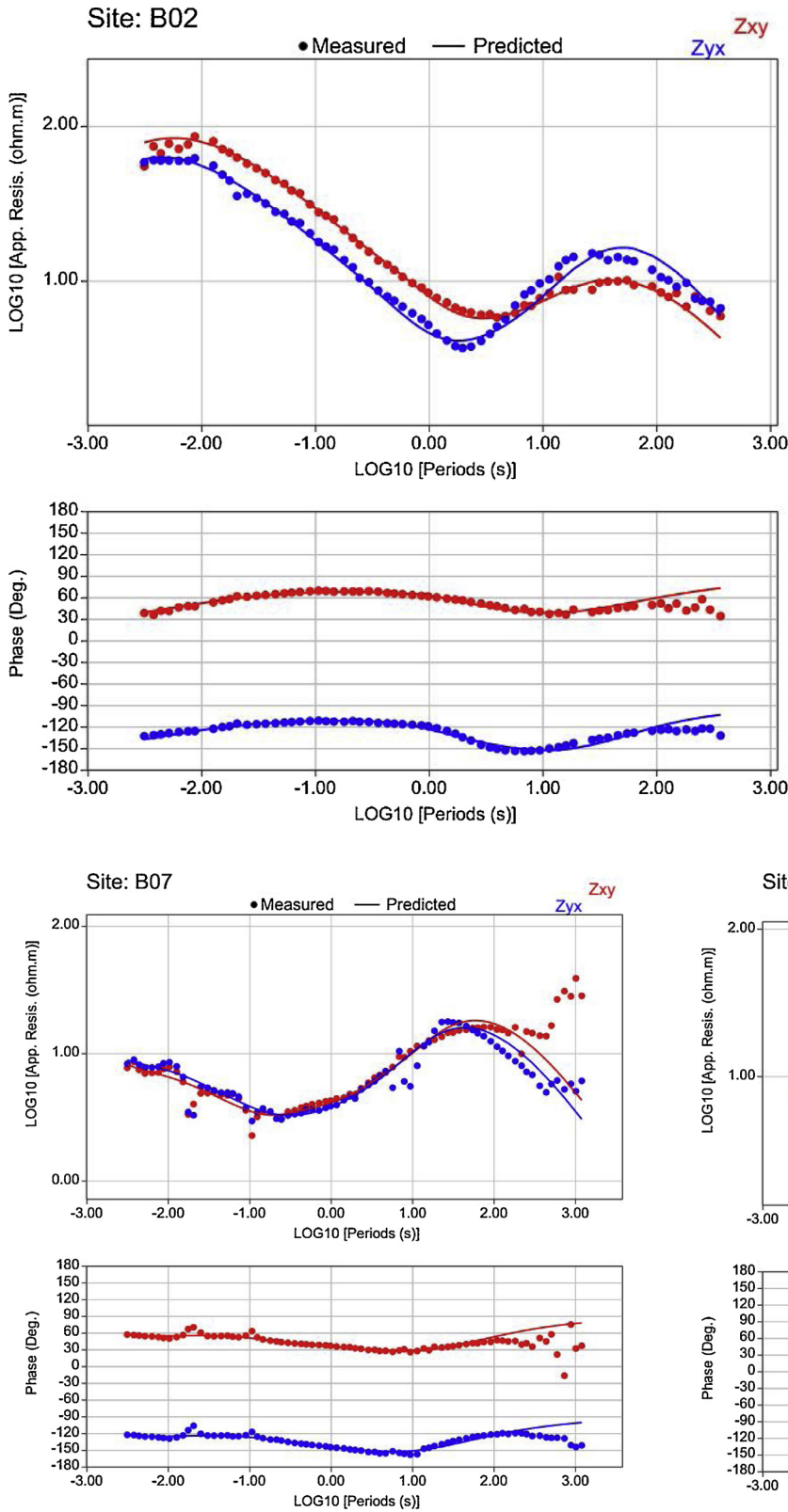


Fig. 16. Data misfit between the observed and calculated apparent resistivity and phase parameters of Zxy (red dots) and Zyx (blue dots) components for station B07. The dots represent the measured values, and the line represents the calculated response. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Fig. 15. Data misfit between the observed and calculated apparent resistivity and phase parameters of Zxy (red dots) and Zyx (blue dots) components for station B02. The dots represent the measured values, and the line represents the calculated response. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

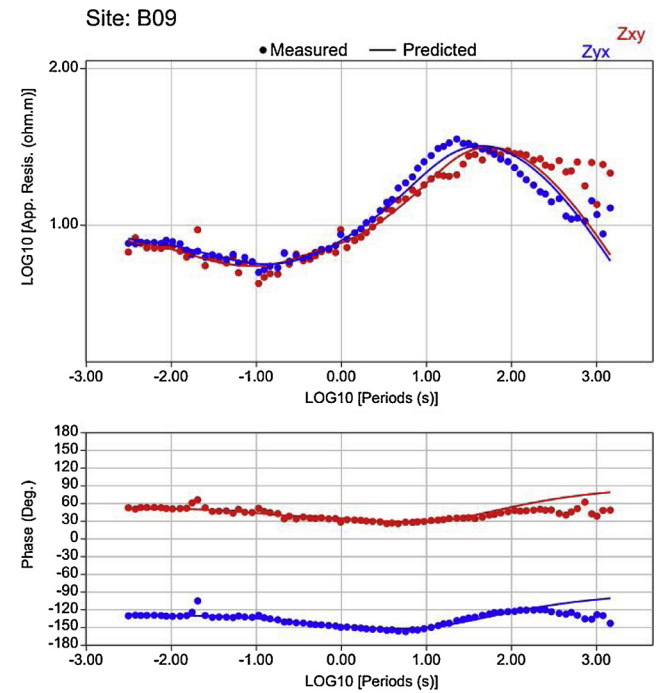


Fig. 17. Data misfit between the observed and calculated apparent resistivity and phase parameters of Zxy (red dots) and Zyx (blue dots) components for station B09. The dots represent the measured values, and the line represents the calculated response. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

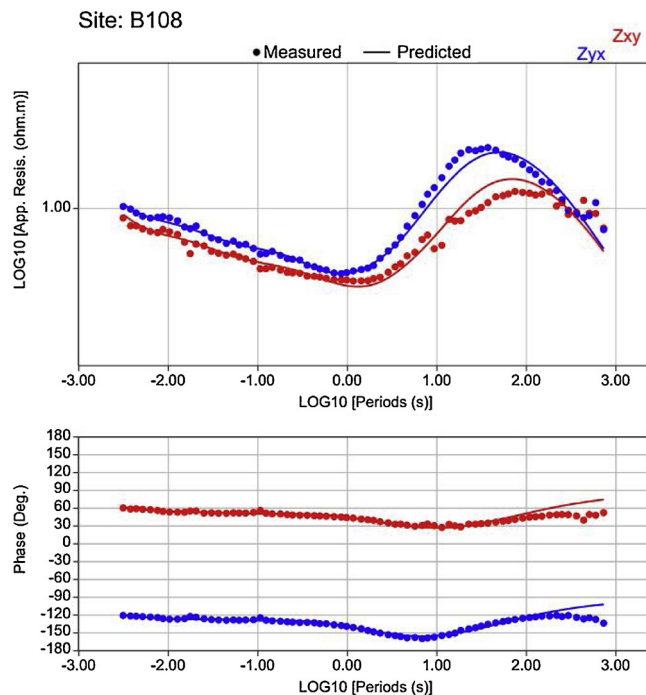


Fig. 18. Data misfit between the observed and calculated apparent resistivity and phase parameters of Zxy (red dots) and Zyx (blue dots) components for station B108. The dots represent the measured values, and the line represents the calculated response. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

D resistivity models indicates that the two models image similar structures. Both 2-D and 3-D models imaged conductors C1, C2 and resistors R1, R2 in profile 1 as shown in Fig. 19. The near-surface layer (C1) is confined to the upper kilometer and has a low resistivity ($< 5 \Omega\text{m}$). Conductor C2 lies below C1, and they appear to be like a connecting conduit that stretches to an elevation of 2.5 km b.s.l. The 3-D resistivity model has a better resolution of the two conductors. In both models, resistor R1 is similar, but in the 2-D model, it is more resistive and extends to a much greater depth. The resistivity of R2 is almost the same in both models, but in 3-D it stretches from the near-surface to an elevation of ~2.5 km b.s.l. However, conductors C1 and C2 are better constrained in the 3-D model since it incorporated all components of the impedance tensor.

Fig. 20 shows the resistivity distribution in Profile 2. In this profile, the two models image conductors C3, C4, and resistor R3 similarly but the 3-D model was able to image conductor C5 on the east side of the profile.

In order to correlate further the resistivity structure with the alteration observed in the wells, a cross-section across well EW-01 was extracted. This cross-section which lies above and parallel to profile 2 indicates the presence of two conductive anomalies, C6 ($\sim 3 \Omega\text{m}$) at an elevation of about 1100 m a.s.l and C7 ($\sim 10 \Omega\text{m}$) starting approximately at sea level and extending to an elevation of 1600 m b.s.l. (Fig. 21). There are corresponding resistivity structures on the western part of the two profiles and the cross-section, but the size and shape differ considerably.

A cross-section cutting through the hot spring (Figs. 2 and 25) and reported geological structures was also extracted in order to further characterize the resistivity distribution of the study area. The profile shows a conductive layer ($\sim 10 \Omega\text{m}$) with variable thickness and depths extending to the surface at soundings B108, B07, B38 and B39 (Fig. 22). This layer overlies a resistive zone that increases from intermediate ($\sim 30 \Omega\text{m}$) to high ($> 100 \Omega\text{m}$) resistivity values and domes beneath the hot spring. The location of the hot spring coincides with the surface termination of a high to low resistivity regions that would suggest the

presence of a fault. There is a consistent high resistivity structure observed on the southern part of the study area as evidenced in the depth slices extending from the near-surface to deeper depths (Figs. 23 and 24). This resistive body, interpreted as a cold formation, appears to surround a high conductive zone. The latter structure seems to delineate a zone of hydrothermal circulation confirmed by the temperatures of drilled geothermal wells EW-01, EW-04 and EW-06 reported in Mwarania (2014).

Fig. 25 shows the iso-resistivity surface ($50 \Omega\text{m}$) that represents a deep resistivity structure below the conductive layer obtained by the inversion. The orientation of this resistive body is S-N direction and correlates with the strike of the local faults system as shown in Fig. 2. The approximate location of the productive wells EW-01, EW-04 and EW-06, marked in Fig. 25 is the southern end of the deep resistivity body.

4. Discussion

In interpreting the resistivity distribution of the model, previous gravity model (Simiyu, 1990) and the drilling and lithologic logs obtained (publication of the logs is not permitted) from the wells were put into consideration. The two resistivity profiles and the cross-section obtained from the 3-D model images seven conductive zones, C1–C7 and three resistors, R1–R3 (Figs. 19–21). The conductors marked as C1, C3 and C6 are highly conductive with values $< 10 \Omega\text{m}$ extending between 600 m–1100 m in depth and presumed to be dominated by low-temperature conductive alteration minerals interpreted to represent cap rock of the geothermal system (Moore et al., 2008). The formation of a clay cap in a geothermal system might be from a liquid at a lower temperature or from steam that condenses at shallow levels creating condensate layers. The clay caps are very conductive and act like a cover that keeps the heat and hot fluids inside the system (Raharjo et al., 2010). Conductor C1 appear to be connected to C2 of Fig. 19, C3 to C4 of Fig. 20 and C6 to C7 of Fig. 21 implying the presence of a deep conducting channel acting as a fluid pathway. Fluids circulating at deeper levels are likely to flow to the surface through the highest permeability pathways (mainly faults). One possible way of transporting heat between the heat source and the reservoir is through the conductive process by means of channels.

Generally, the tectonic activity acting in the Kenya rift system manifested itself locally by the formation of numerous buried faults striking in the S-N direction in Eburru. These faults are associated with low resistivity anomalies which can be interpreted as the conduits transporting thermal energy from the heat source to shallow depths (Wannamaker et al., 2006). Therefore, the deep-seated low resistivity anomalies (C2, C4 and C7) could be due to magmatic fluid components arising from deeper melt sources responsible for heating up the resistive reservoir. The spatial distribution of these anomalies (C2, C4 and C7), with high conductivities on the western part, is consistent with the location of fumarolic activity in the area (Fig. 10). It is known that partial melting is a conductive material which could be responsible for these conductive anomalies (Peacock et al., 2016; Wannamaker et al., 2008). The resistive structures R2 (Fig. 19) and R3 (Fig. 20) were interpreted to represent impermeable or cold formations.

The shallow layer consists mainly of fresh volcanic materials like tephra, pyroclastic rock or tuff with formation temperature generally $< 70^\circ\text{C}$ estimated from the drilled wells (Samrock et al., 2015). In addition, downhole pressure measurements indicate the water table is at a depth of about 600 m and therefore above this depth the water saturation is generally low. Such low temperatures within the unsaturated zone are unlikely to cause any significant hydrothermal alteration and hence results in high resistivities (Ussher et al., 2000). The temperature profiles of well EW-01 generally shows low measured values (less than 170°C) above the depths of about 450 m. The well appears to penetrate the clay cap region (conductor C6 $< 10 \Omega\text{m}$) at a depth of 1000 m where the temperature rises to 210°C . This region is

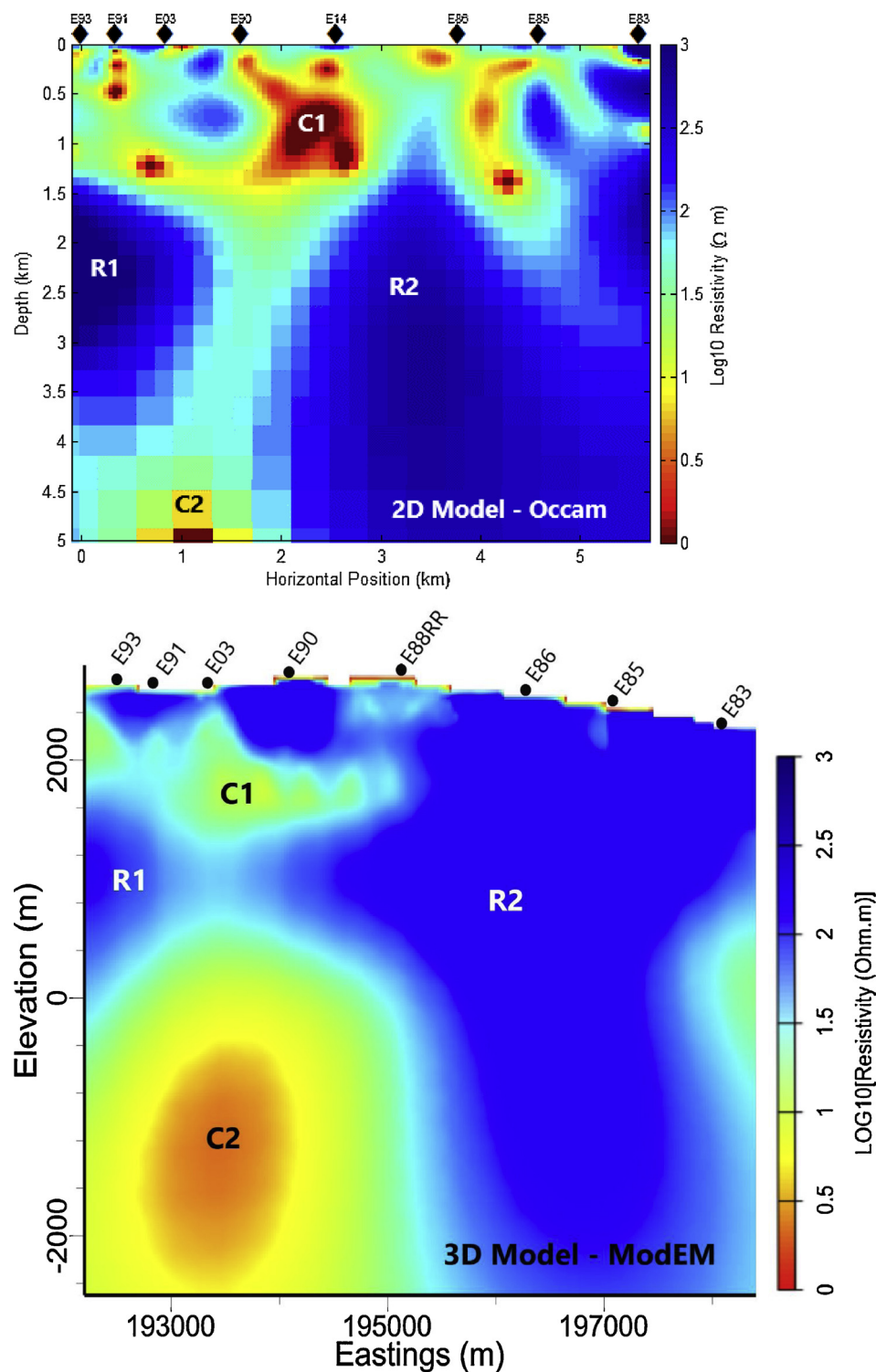


Fig. 19. 2-D and 3-D resistivity models for Profile 1.

usually characterized by the formation of illites and smectites, where smectite content rapidly decreases as it is progressively converted to illite with an increase in temperature (Johnston et al., 1992). Beneath this region is the propylitic zone where hot geothermal fluids, migrate upwards along fractured formations and narrow faults, which was encountered at a temperature of 238 °C and a depth of about 1100 m. Well EW-04 shown in Figs. 12 and 20 has a total depth of 2469 m and penetrates the clay cap (conductor C3 < 10 Ωm) at a depth of 1000 m from the surface, underlain by the propylitic zone (15–60 Ωm). The clay

cap is quite extensive and stretches down to a thickness of about 500 m. Beyond this region, we find the propylitic reservoir that has been proven by drilling, and from the resistivity structure, this zone is interpreted to be on the southwest of the study area. A loss of circulation of approximately 150 m thick was penetrated by the well, which suggest a possibility of a loose formation or fractured zone.

The fault-controlled geometry of the deep resistive body of Fig. 25 is most probably associated with the flow pattern of the geothermal fluid in Eburru geothermal field. The productive wells dip towards this

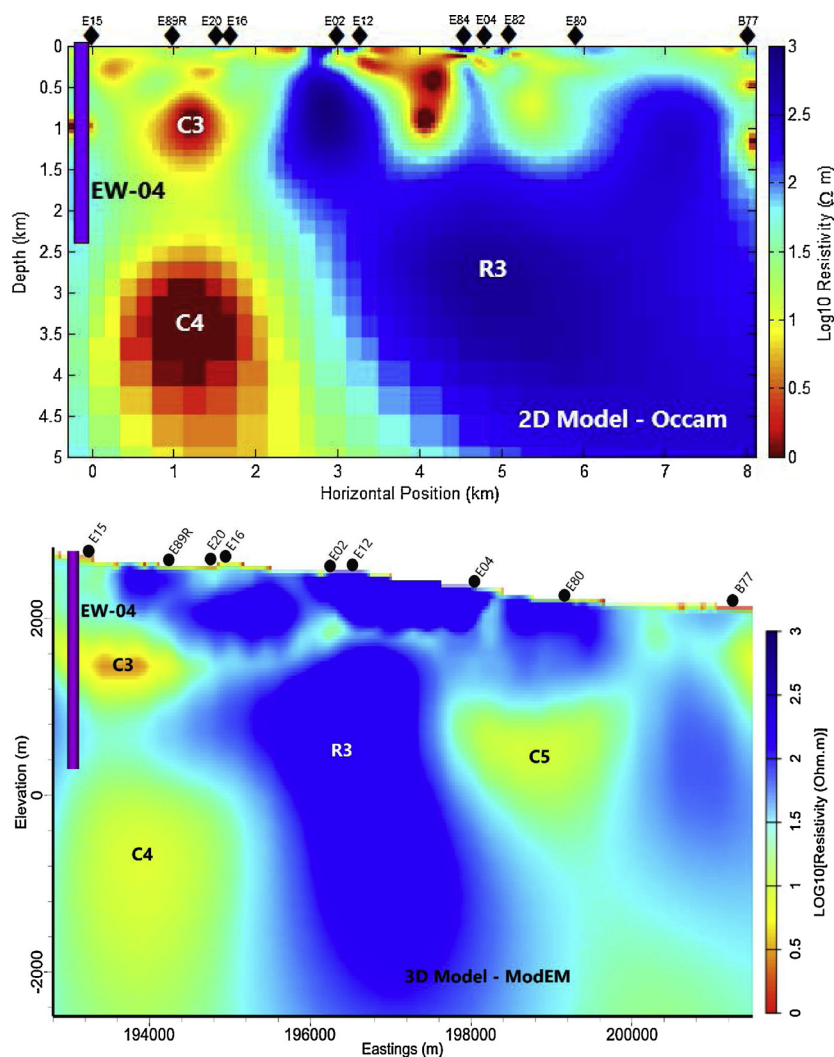


Fig. 20. 2-D and 3-D resistivity models for Profile 2.

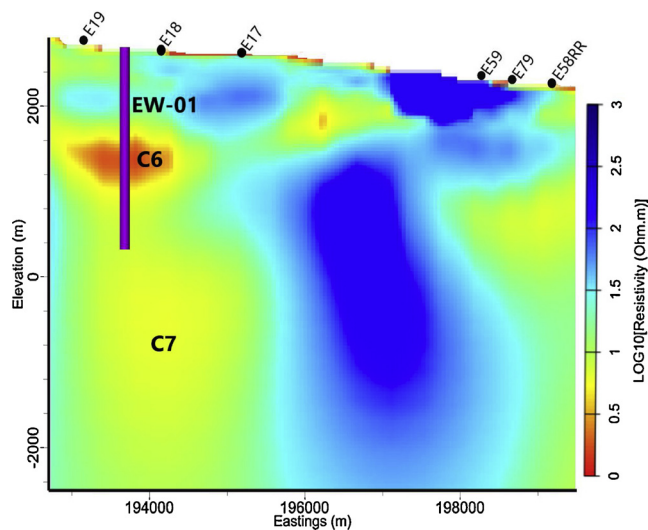


Fig. 21. 3-D resistivity model cutting through well EW-01.

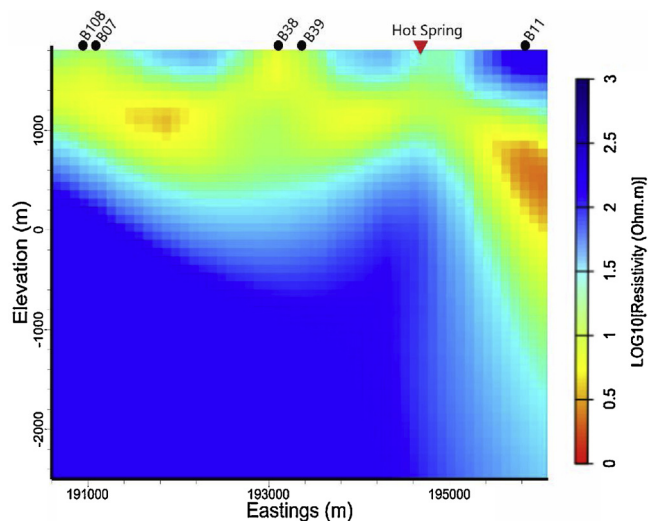


Fig. 22. 3-D resistivity model cutting through hot spring.

structure suggesting some correlation with high permeability and fluid content. The *iso*-surface of 50 Ωm appears to surround a conductive zone towards the southern end likely to be due to magmatic fluid components and is interpreted as the possible heat source of the

geothermal system. The productive wells appear to be tapping from a reservoir which overlies this conductive zone. This interpretation is supported by gravity data which shows that the high-density intrusive bodies occur along these fault zones (Simiyu, 1990). This inference can

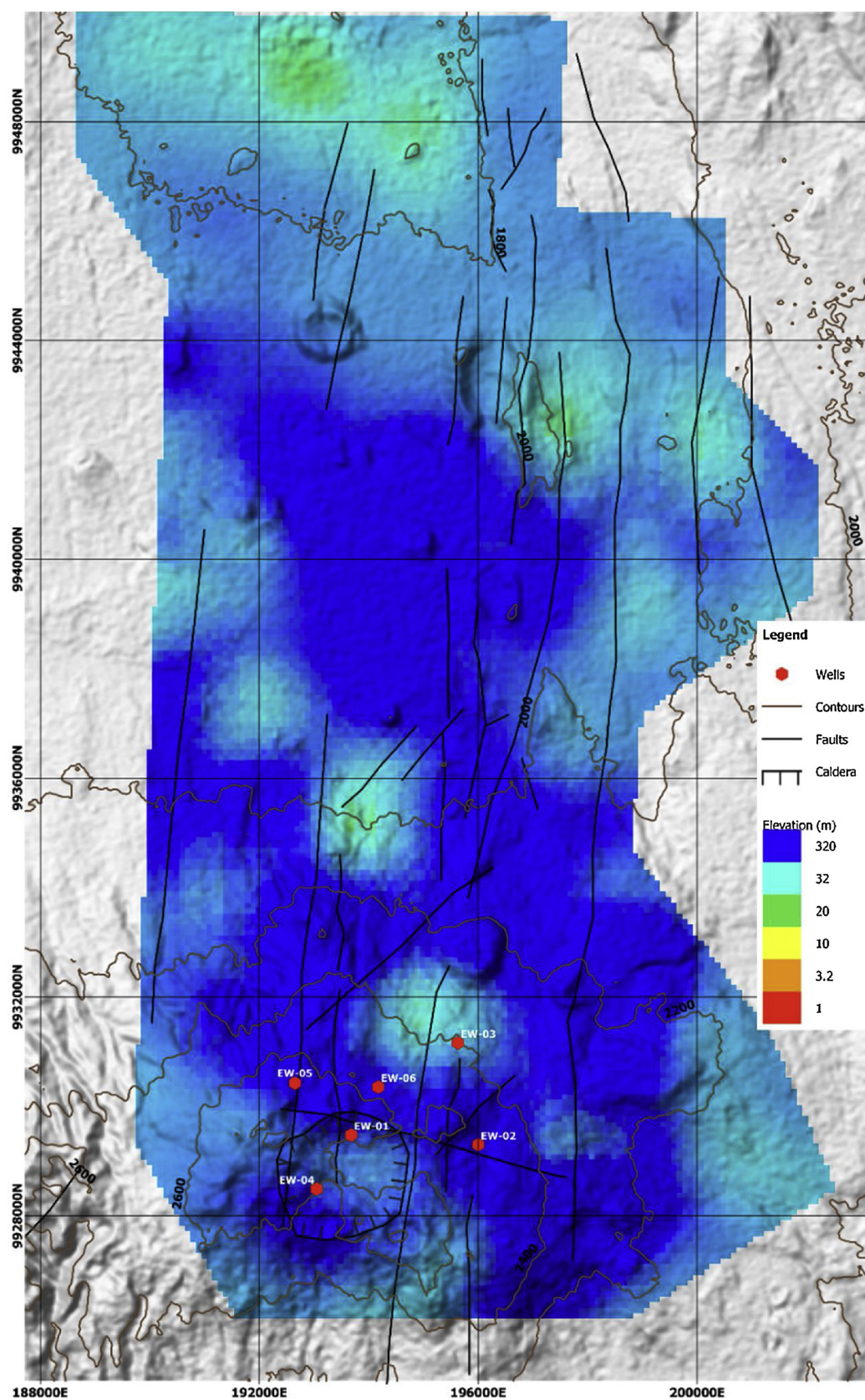


Fig. 23. Planar view at an elevation of 1700 m of the resistivity distribution.

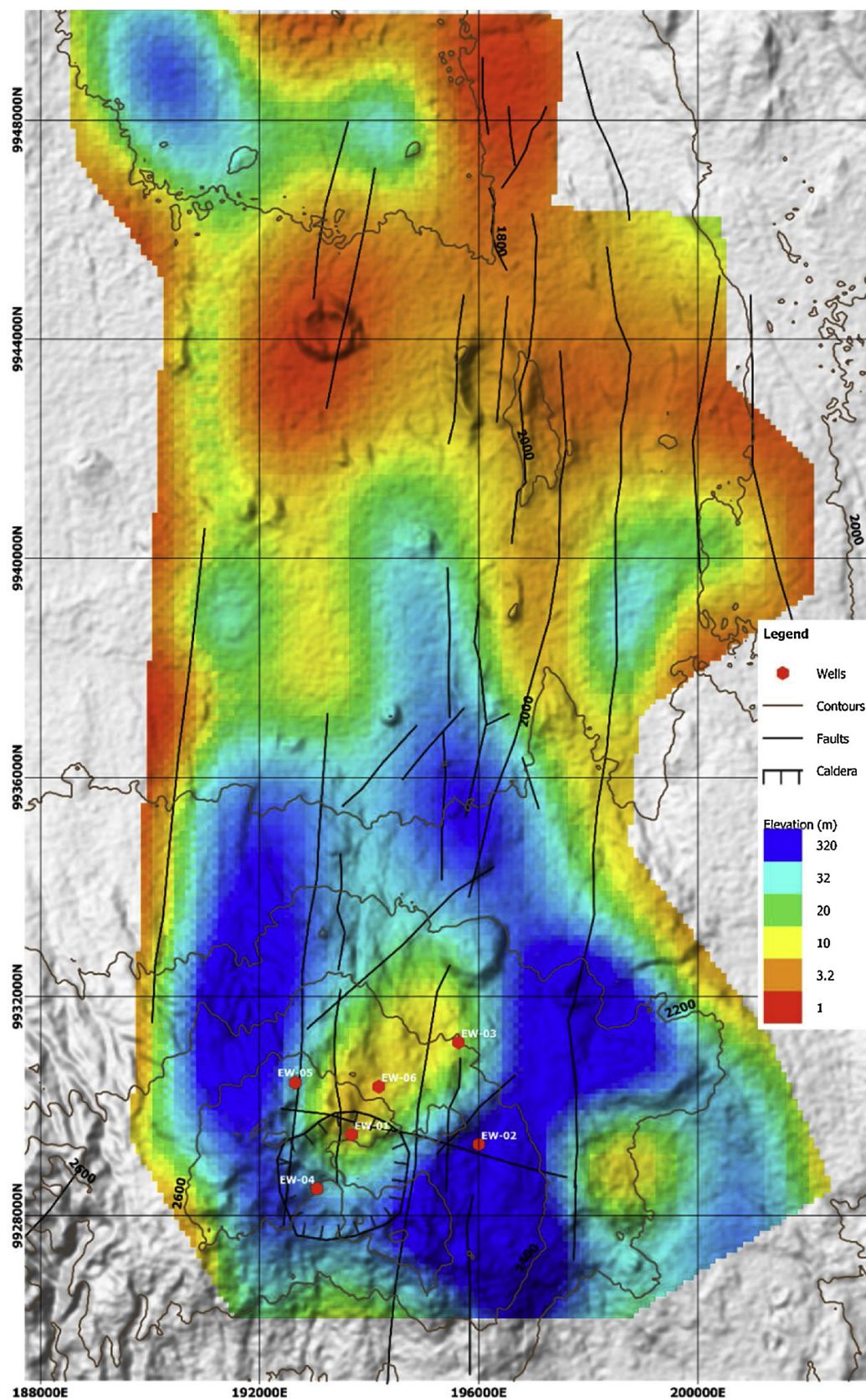


Fig. 24. Planar view at an elevation of 250 m of the resistivity distribution.

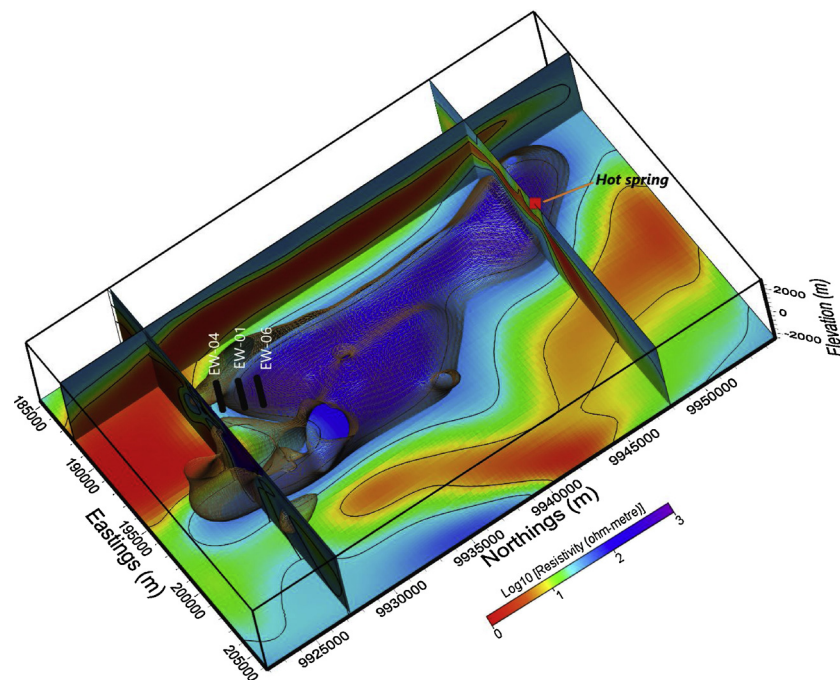


Fig. 25. 3-D resistivity model of the Eburru geothermal area showing an isosurface of 50 Ωm with planar views extracted from the model.

be ascertained by inputs of other geophysical methods such as seismic and intense gravity or by further exploratory drilling.

5. Conclusion

In this study, both 2-D and 3-D inversions were carried out, and the interpretation of the results made based on these two types of resistivity model. Resistivity structures obtained from the two inversions were imaged similarly. The two models recovered a conductive zone ($< 10 \Omega\text{m}$) within 1 km from the surface overlaying a relatively resistive layer ($\sim 35 \Omega\text{m}$) and a deep conductor ($< 10 \Omega\text{m}$) extending from sea level to a depth of 2.5 km below sea level. The profiles analyzed revealed a structure which can be interpreted as the potential pathway used by the geothermal fluid to migrate to the surface. The local faults trending in the S-N direction in the field which serves as transporting channels of heat from deep sources to the geothermal reservoir were imaged as low resistivity anomalies. The deeper low resistivity anomalies may be as a result of magmatic fluid components originating from deeper melt sources.

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